GCSE BIOLOGY

Higher Tier – Paper 1 style

Topic 4.2: Organisation

Practice questions in the style of AQA GCSE Biology (8461). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 235 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **224**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 224

0 1 This question is about how living organisms are organised.

0 1 . 1 Complete the sentences. Choose words from the box. **[2 marks]**

A group of similar cells make up _____ .

Several of these joined together form an _____ .

cells	**organs**	**organ**	**tissues**	**organ systems**

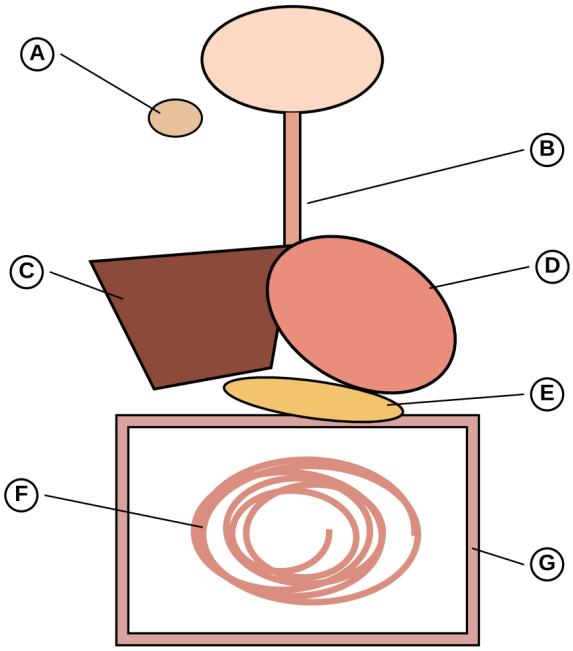
0 1 . 2 Put these levels of organisation in order, starting with the smallest. **[2 marks]**

organ system cell organism tissue organ

0 1 . 3 The stomach is an organ. Give **two** tissues found in the stomach and state the function of each. **[4 marks]**

0 1 . 4 Name the organ system that contains the stomach, the liver and the pancreas. **[1 mark]**

0 2 The diagram shows part of the human digestive system. The letters **A** to **G** show different organs.



0 2 . 1 Name the organs labelled **A** to **G**. **[7 marks]**

0 2 . 2 Which organ produces **bile**? Choose a letter. **[1 mark]**

0 2 . 3 Which organ makes insulin and also produces digestive enzymes? Choose a letter. **[1 mark]**

0 2 . 4 Describe what happens in organ **F**. **[2 marks]**

0 2 . 5 Water is absorbed in organ **G**. Explain why the contents of G are more solid than the contents of F. **[2 marks]**

0 2 . 6 Explain why digestion is necessary. **[2 marks]**

0 3 Enzymes are biological catalysts. The diagrams show an enzyme and its substrate. Diagrams **A** to **D** show four stages in an enzyme-catalysed reaction.



0 3 . 1 What is meant by a **catalyst**? **[1 mark]**

0 3 . 2 Which diagram shows the enzyme–substrate complex? Choose a letter. **[1 mark]**

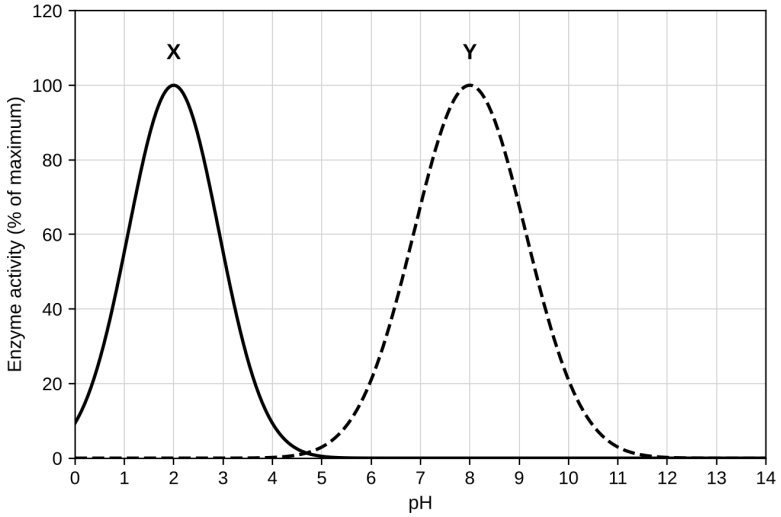
0 3 . 3 Explain why enzymes are **specific** for one substrate. **[2 marks]**

0 3 . 4 The enzyme in diagram **D** has been heated to 80 °C. Describe what has happened to the enzyme and explain why the substrate no longer fits. **[3 marks]**

0 3 . 5 Suggest why an enzyme can be used again after the reaction in diagram **C**. **[1 mark]**

0 3 . 6 The rate of an enzyme-catalysed reaction increases between 10 °C and 40 °C. Explain why. **[2 marks]**

0 4 The graph shows how the activity of two enzymes, **X** and **Y**, changes with pH. One enzyme is a protease from the stomach. The other is a lipase from the small intestine.



0 4 . 1 State the optimum pH of enzyme **X** and of enzyme **Y**. **[2 marks]**

0 4 . 2 Which enzyme, **X** or **Y**, is the protease from the stomach? Give a reason for your answer. **[2 marks]**

0 4 . 3 Use the graph to find the activity of enzyme **Y** at pH 6. **[1 mark]**

0 4 . 4 Enzyme **Y** has almost no activity at pH 2. Explain why. **[2 marks]**

0 4 . 5 Explain how the small intestine is kept at a pH suitable for enzyme **Y** even though food from the stomach is acidic. **[2 marks]**

0 5 Digestive enzymes break down large food molecules into small molecules.

0 5 . 1 Complete the table. **[5 marks]**

Enzyme	Substrate	Product(s)
	starch	maltose / sugars
protease		amino acids
	fats (lipids)	

0 5 . 2 Name **two** places where amylase is produced. **[2 marks]**

0 5 . 3 Maltose is broken down into glucose in the small intestine. Name the type of enzyme that does this. **[1 mark]**

0 5 . 4 Describe the functions of bile in digestion. **[3 marks]**

0 5 . 5 Explain why emulsifying fat helps lipase to work faster. **[2 marks]**

0 6 A student tests four food samples, **P**, **Q**, **R** and **S**, using different food tests.
The table shows her results.

0 6 . 1 Complete the table to show the food tests and the positive results. **[4 marks]**

Food molecule	**Reagent**	**Positive result**
sugar (reducing)		
starch		
protein	biuret	purple / lilac
lipid	ethanol (then water)	cloudy white emulsion

0 6 . 2 The student tests the samples with **two** tests. Her results are shown. **[2 marks]**
Sample **P**: iodine stays orange-brown; Benedict's stays blue.
Sample **Q**: iodine turns blue-black.
Sample **R**: biuret turns purple.
Sample **S**: Benedict's turns brick red.
Which sample contains **starch**? Which contains a **reducing sugar**?

0 6 . 3 Which sample contains protein? Give a reason. **[1 mark]**

0 6 . 4 The Benedict's test needs heat. Describe how the student should safely heat the mixture. **[2 marks]**

0 6 . 5 Describe how to test a solid food sample for lipids. **[3 marks]**

0 7 A student investigates the effect of pH on the activity of amylase. She mixes starch solution, amylase and a buffer at 35 °C. Every 30 seconds she takes a drop of the mixture and adds it to iodine solution on a spotting tile. She records the time taken for the iodine to stop turning blue-black. The table shows her results.

0 7 . 1 Complete the table by calculating the rate of reaction at pH 7. **[2 marks]**
Rate = 1000 ÷ time (arbitrary units).

pH of buffer	**4**	**5**	**6**	**7**	**8**
Time for blue-black colour to disappear in s	240	110	40	60	180
Rate (1000 ÷ time) in a.u.	4.2	9.1	25.0		5.6

Answer = _____ a.u.

0 7 . 2 What is the optimum pH for the amylase of those tested? Give a reason. **[2 marks]**

0 7 . 3 Suggest how the student could find the optimum pH more precisely. **[1 mark]**

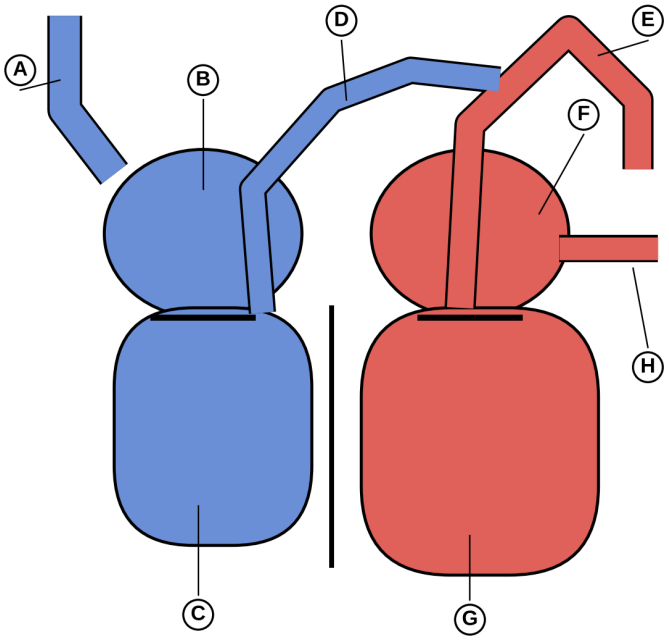
0 7 . 4 Name the **independent variable** and the **dependent variable** in this investigation. **[2 marks]**

0 7 . 5 Give **two** variables the student must control. **[2 marks]**

0 7 . 6 A drop of the mixture is added to iodine every 30 seconds. Suggest why 30 seconds is a suitable interval for pH 6 but not for pH 7. **[2 marks]**

0 7 . 7 Explain why the amylase has almost no activity at pH 4. **[2 marks]**

0 8 The diagram shows the human heart and the blood vessels attached to it. Chambers and vessels are labelled **A** to **H**. The heart is part of the circulatory system.



0 8 . 1 Name the parts labelled **A** to **H**. **[8 marks]**

0 8 . 2 Which labelled blood vessel carries **oxygenated** blood from the heart to the body? Choose a letter. **[1 mark]**

0 8 . 3 The wall of the left ventricle is much thicker than the wall of the right ventricle. Explain why. **[2 marks]**

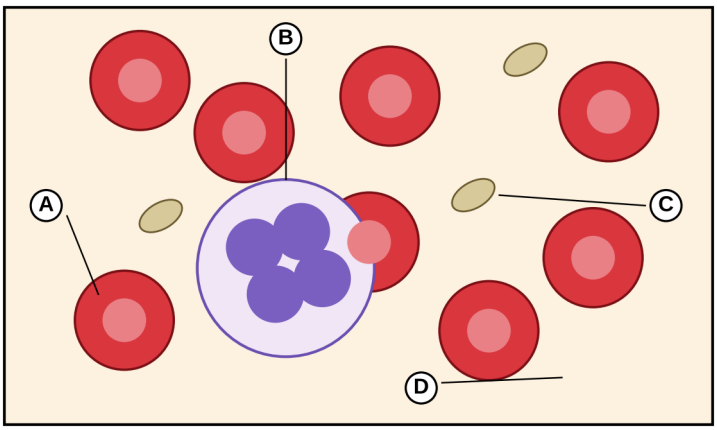
0 8 . 4 The heart has valves. Describe the function of the valves. **[2 marks]**

0 8 . 5 Describe the path taken by blood from the vena cava to the lungs. **[3 marks]**

0 8 . 6 The human circulatory system is a **double circulatory system**. Explain the advantage of this. **[2 marks]**

0 8 . 7 The natural resting heart rate is controlled by a group of cells in the right atrium. Name these cells and state what happens if they do not work properly. **[2 marks]**

0 9 The diagram shows some of the components of human blood as seen under a light microscope.



0 9 . 1 Name the components labelled **A**, **B**, **C** and **D**. **[4 marks]**

0 9 . 2 Explain how **A** is adapted to its function. **[3 marks]**

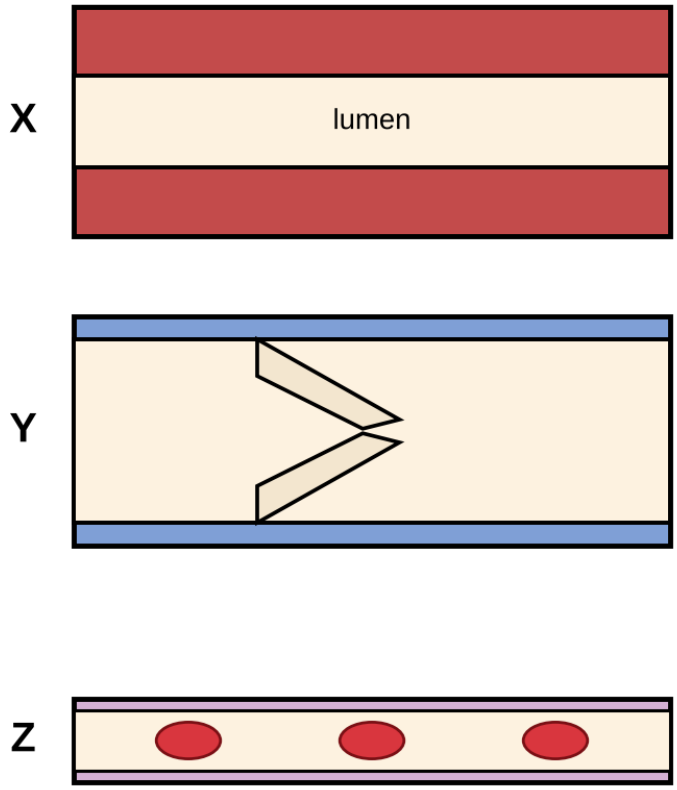
0 9 . 3 Describe **two** ways in which **B** protects the body against pathogens. **[2 marks]**

0 9 . 4 Describe the function of **C**. **[2 marks]**

0 9 . 5 Name **three** substances transported by **D**. **[3 marks]**

0 9 . 6 A person living at high altitude has more red blood cells than a person living at sea level. Explain why this helps them. **[2 marks]**

1 0 The diagram shows three types of blood vessel, **X**, **Y** and **Z**.



1 0 . 1 Name the blood vessels **X**, **Y** and **Z**. **[3 marks]**

1 0 . 2 Explain how the structure of **X** is related to its function. **[3 marks]**

1 0 . 3 Vessel **Y** has valves. Explain why. **[2 marks]**

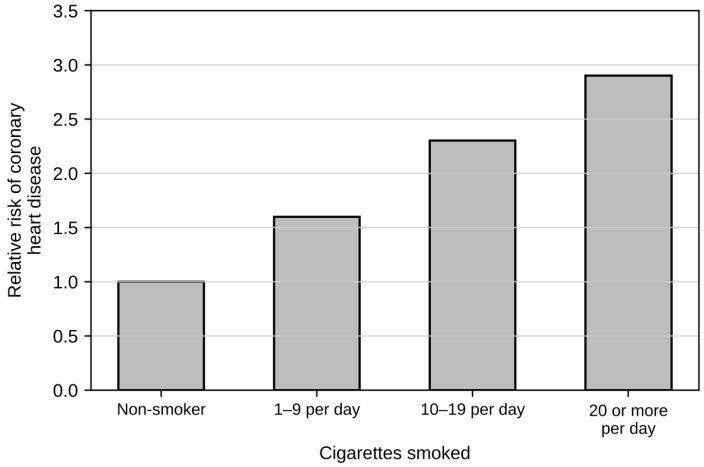
1 0 . 4 Vessel **Z** has walls that are only one cell thick. Explain why this is important. **[2 marks]**

1 0 . 5 A pulse can be felt at an artery but not at a vein. Explain why. **[2 marks]**

1 0 . 6 The blood in a capillary travels at 0.4 mm per second. Calculate how long it takes for blood to travel 2.0 mm along the capillary. **[2 marks]**

Answer = _____ s

1 1 Coronary heart disease (CHD) is a non-communicable disease. The bar chart shows how the risk of developing CHD changes with the number of cigarettes smoked per day. The risk for a non-smoker is given a value of 1.0.



1 1 . 1 What is meant by a **non-communicable disease**? **[1 mark]**

1 1 . 2 Describe the effect of smoking on the risk of CHD. **[2 marks]**

1 1 . 3 Calculate the percentage increase in risk for a person who smokes 20 or more cigarettes per day compared with a non-smoker. **[3 marks]**

Answer = _____ %

1 1 . 4 Explain how coronary heart disease can lead to a heart attack. **[3 marks]**

1 1 . 5 Statins reduce the risk of CHD. Give **one** advantage and **one** disadvantage of taking statins. **[2 marks]**

1 1 . 6 A stent is used to treat a blocked coronary artery. Describe how a stent helps. **[2 marks]**

1 1 . 7 The graph shows a correlation. A student concludes that smoking causes CHD. Explain whether the graph alone proves this. **[2 marks]**

1 2 This question is about health and disease.

1 2 . 1 Many diseases are caused by a combination of factors. Describe how **two** different diseases can interact. Give an example. **[2 marks]**

1 2 . 2 The table shows the results of a study of 4000 adults. It shows how many developed type 2 diabetes. **[2 marks]**

Calculate the percentage of obese adults who developed type 2 diabetes.

Group	Number of adults	Number who developed type 2 diabetes
obese	1500	180
not obese	2500	60

Answer = _____ %

1 2 . 3 Calculate how many times greater the percentage is in obese adults than in adults who are not obese. **[3 marks]**

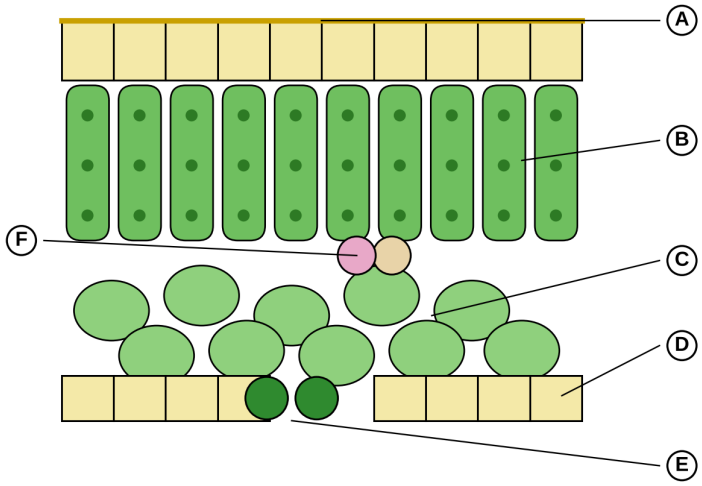
Answer = _____ times

1 2 . 4 Name **two** other risk factors for non-communicable diseases and the disease each one increases the risk of. **[4 marks]**

1 2 . 5 Describe the difference between a **benign** tumour and a **malignant** tumour. **[3 marks]**

1 2 . 6 Suggest **two** ways in which non-communicable diseases affect (a) a person, and (b) the country's health service. **[4 marks]**

1 3 The diagram shows a section through a leaf. The letters **A** to **F** show different parts.



1 3 . 1 Name the parts labelled **A** to **F**. **[6 marks]**

1 3 . 2 Explain how part **B** is adapted for photosynthesis. **[3 marks]**

1 3 . 3 Explain how part **C** allows gas exchange. **[2 marks]**

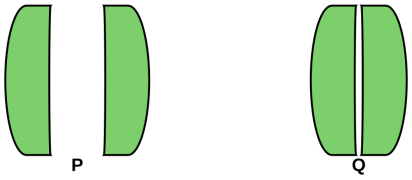
1 3 . 4 Explain why the upper epidermis is transparent. **[1 mark]**

1 3 . 5 Name the tissue that carries water in a leaf and state **one** way it is adapted. **[2 marks]**

1 4 The diagrams show a stoma in a leaf when it is open (**P**) and when it is closed (**Q**).

1 4 . 1 Name the cells that open and close a stoma. **[1 mark]**

1 4 . 2 Explain how the stoma changes from **Q** to **P**. **[2 marks]**



1 4 . 3 Explain why the stomata are usually closed at night. **[2 marks]**

1 4 . 4 Define **transpiration**. **[2 marks]**

1 4 . 5 A student measures the rate of transpiration using a potometer. The air bubble moves 36 mm in 6 minutes. The capillary tube has a radius of 0.5 mm. **[4 marks]**

Calculate the volume of water lost per minute. (Volume = $\pi r^2 \times$ distance, $\pi = 3.14$)

Answer = _____ mm^3/min

1 4 . 6 Explain how **each** of the following affects the rate of transpiration. **[4 marks]**

(a) increased light intensity

(b) increased humidity

1 4 . 7 Describe the movement of water from the soil to the air through a plant. **[3 marks]**

1 5 Plants transport substances in two types of tissue: xylem and phloem.

1 5 . 1 Complete the table to compare xylem and phloem. **[4 marks]**

	Xylem	**Phloem**
Substances carried		
Type of cells		

1 5 . 2 What is meant by **translocation**? **[2 marks]**

1 5 . 3 Suggest why the phloem moves sucrose in **both** directions in a plant, unlike the xylem. **[2 marks]**

1 5 . 4 A student removes a ring of bark and phloem from a woody stem. After some weeks the stem swells **above** the ring. Explain why. **[2 marks]**

1 6 Stretch questions (grade 7–9).

1 6 . 1 *Describe how the structure of the heart and blood vessels makes the double circulatory system efficient. Explain the importance of the valves and the different thicknesses of the ventricle walls. **[6 marks]**

1 6 . 2 *A scientist says that digestion in the small intestine depends on the stomach, the liver and the pancreas. Explain why the small intestine can only work properly with help from these organs. **[6 marks]**

1 6 . 3 *Evaluate the treatments available for a patient with coronary heart disease: statins, a stent and a heart transplant. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.2: Organisation

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	tissues ; organ ;	One mark each.	2	AO1 4.2.1
01.2	cell, tissue, organ, organ system, organism ; (any four in the correct consecutive order for 1 mark) ;		2	AO1 4.2.1
01.3	muscular tissue ; churns / mixes the food with the digestive juices ; glandular tissue ; makes / releases digestive juices (acid, protease enzymes) ;	Allow epithelial tissue (covers the inside and outside of the organ) for 1 mark + its function for 1 mark. Any two tissues.	4	AO1 4.2.1
01.4	digestive (system)		1	AO1 4.2.1
02.1	A: salivary gland ; B: oesophagus / gullet ; C: liver ; D: stomach ; E: pancreas ; F: small intestine ; G: large intestine ;	One mark for each correct name.	7	AO1 4.2.2.1
02.2	C		1	AO1 4.2.2.1
02.3	E		1	AO1 4.2.2.1
02.4	digestion is completed (by enzymes) ; soluble food molecules are absorbed into the blood ;		2	AO1 4.2.2.1
02.5	water is absorbed from the undigested food ; in G (so the remains / faeces become more solid) ;		2	AO2 4.2.2.1
02.6	food molecules (starch, proteins, fats) are too large / insoluble to be absorbed ; digestion breaks them into small soluble molecules that pass into the blood ;		2	AO2 4.2.2.1
03.1	a substance that speeds up a reaction without being used up / changed		1	AO1 4.2.2.2
03.2	B		1	AO2 4.2.2.2
03.3	the (shape of the) active site matches / is complementary to the substrate ; only that substrate fits into the active site (lock and key model) ;		2	AO1 4.2.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	the enzyme has been denatured ; the shape of the active site has changed ; so the substrate no longer fits / cannot bind ;	Allow 'bonds holding the protein break'.	3	AO2 4.2.2.2
03.5	the enzyme is not used up / is unchanged by the reaction		1	AO2 4.2.2.2
03.6	particles / molecules have more energy and move faster ; so there are more collisions between enzyme and substrate (more successful collisions) ;		2	AO2 4.2.2.2
04.1	X: pH 2 ; Y: pH 8 ;		2	AO3 4.2.2.2
04.2	X ; the stomach contains hydrochloric acid / X works best in acidic conditions (pH 2) ;		2	AO3 4.2.2.2
04.3	19–23 (%)		1	AO3 WS 3.1
04.4	the acid / low pH denatures the enzyme ; the shape of the active site changes so the substrate no longer fits ;		2	AO2 4.2.2.2
04.5	bile (made in the liver) is alkaline ; it neutralises the hydrochloric acid from the stomach ;		2	AO2 4.2.2.3
05.1	amylase ; starch ; amino acids ; lipase ; fatty acids and glycerol ;	Amylase → maltose (1); protease substrate = protein (1); lipase (1); products fatty acids and glycerol (2 for both, 1 for one). Max 5.	5	AO1 4.2.2.2
05.2	salivary glands ; pancreas / small intestine ;		2	AO1 4.2.2.2
05.3	carbohydrase / maltase	Do not accept amylase.	1	AO1 4.2.2.2
05.4	bile is alkaline and neutralises acid from the stomach ; to give the best / optimum pH for the enzymes in the small intestine ; bile emulsifies fat (breaks into small droplets) ;	Maximum 3 from these ideas.	3	AO1 4.2.2.3
05.5	emulsification increases the surface area of the fat ; so lipase can break down the fat more quickly ;		2	AO2 4.2.2.3
06.1	Benedict's (solution) ; brick red / orange / green (precipitate) ; iodine (solution) ; blue-black ;	Biuret and lipid rows are given. Allow 'blue to brick-red' for the Benedict's result; 'orange / browny-orange to blue-black' for iodine.	4	AO1 4.2.2.1 / RP
06.2	Q (starch) ; S (reducing sugar) ;		2	AO3 4.2.2.1 / RP
06.3	R as the biuret test is positive		1	AO3 4.2.2.1 / RP

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	place the test tube in a water bath at about 80 °C / hot water bath ; for a few minutes (wear eye protection / use a test-tube holder) ;		2	AO1 4.2.2.1 / RP
06.5	grind / mix the food with ethanol ; filter / let the solid settle and pour the liquid into water ; a cloudy white emulsion means lipid is present ;		3	AO1 4.2.2.1 / RP
07.1	1000 ÷ 60 ; 16.7 (allow 17) ;		2	AO2 4.2.2.2 / RP
07.2	pH 6 ; it has the highest rate / shortest time ;		2	AO3 4.2.2.2 / RP
07.3	repeat with pH values between 5 and 7 (e.g. 5.5, 6.5) / smaller pH intervals		1	AO3 4.2.2.2 / RP
07.4	independent: pH ; dependent: time for the starch to be broken down / rate of reaction ;		2	AO3 WS 2.3
07.5	temperature ; concentration / volume of amylase (or starch) ;	Allow: volume of buffer; time between samples. Any two.	2	AO3 WS 2.3
07.6	pH 6 takes only 40 s so 30 s gives few readings / 30 s is too long to be precise ; so smaller intervals (e.g. 10 s) would give more accurate times for fast reactions ;	Accept any sensible idea about precision.	2	AO3 WS 3.4
07.7	acidic pH denatures the enzyme ; active site changes shape so starch cannot bind ;		2	AO2 4.2.2.2
08.1	A: vena cava ; B: right atrium ; C: right ventricle ; D: pulmonary artery ; E: aorta ; F: left atrium ; G: left ventricle ; H: pulmonary vein ;	One mark for each correct name.	8	AO1 4.2.2.4
08.2	E		1	AO2 4.2.2.4
08.3	the left ventricle pumps blood around the whole body ; needs more force / higher pressure than pumping to the lungs ;		2	AO2 4.2.2.4
08.4	to prevent the backflow of blood ; so blood flows in one direction only ;		2	AO1 4.2.2.4
08.5	vena cava → right atrium ; right atrium → right ventricle ; right ventricle → pulmonary artery → lungs ;		3	AO1 4.2.2.4
08.6	blood returns to the heart to be pumped again after the lungs ; so the blood is at a higher pressure / flows faster to the body (the lungs need a lower pressure) ;		2	AO2 4.2.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.7	pacemaker (cells) ; an artificial pacemaker is fitted to correct the heart rate / irregular heartbeat ;		2	AO1 4.2.2.4
09.1	A: red blood cell ; B: white blood cell ; C: platelet(s) ; D: plasma ;		4	AO1 4.2.2.5
09.2	contains haemoglobin which binds / carries oxygen ; no nucleus so more room for haemoglobin ; biconcave shape gives a large surface area for oxygen absorption ;		3	AO2 4.2.2.5
09.3	engulfs / digests pathogens (phagocytosis) ; produces antibodies / antitoxins ;		2	AO1 4.2.2.5
09.4	helps blood to clot ; at a wound to stop bleeding / stop pathogens getting in ;		2	AO1 4.2.2.5
09.5	carbon dioxide ; urea ; glucose / hormones / digested food ;	Allow: amino acids, hormones, antibodies, heat. Any three.	3	AO1 4.2.2.5
09.6	there is less oxygen in the air at high altitude ; more red blood cells can carry more oxygen to the tissues ;		2	AO3 4.2.2.5
10.1	X: artery ; Y: vein ; Z: capillary ;		3	AO1 4.2.2.5
10.2	carries blood away from the heart at high pressure ; thick muscular walls ; elastic fibres allow it to stretch and recoil ;		3	AO2 4.2.2.5
10.3	blood in veins is at low pressure ; valves prevent backflow of blood ;		2	AO2 4.2.2.5
10.4	short diffusion distance ; so substances (oxygen, glucose) pass quickly between blood and tissues ;		2	AO2 4.2.2.5
10.5	the heart pumps blood into arteries in surges / at high pressure ; pressure in veins is low and steady ;		2	AO3 4.2.2.5
10.6	time = distance ÷ speed ; 5 seconds ;		2	AO2 MS 3b
11.1	a disease that cannot be passed from one person to another / not caused by pathogens		1	AO1 4.2.2.6
11.2	the more cigarettes smoked, the higher the risk ; risk rises from 1.0 to 2.9 (data quote) ;		2	AO3 4.2.2.6 / WS 3.5
11.3	2.9 − 1.0 = 1.9 ; 1.9 ÷ 1.0 × 100 ; 190 % ;		3	AO2 MS 1c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
11.4	fatty material builds up in the coronary arteries ; narrowing the arteries and reducing blood flow to the heart muscle ; less oxygen / glucose reaches the muscle so it cannot respire / contract ;		3	AO1 4.2.2.6
11.5	advantage: reduces blood cholesterol / slows fatty build-up ; disadvantage: may have side effects (e.g. muscle pain, liver problems) / must be taken for life ;		2	AO1 4.2.2.6
11.6	a stent keeps the artery open ; so blood can flow to the heart muscle ;		2	AO1 4.2.2.6
11.7	no, a correlation does not prove a cause ; other factors (diet, genes, exercise) could also be linked / need evidence about the mechanism ;		2	AO3 WS 1.4 / 3.5
12.1	defects in the immune system make an individual more likely to suffer from infectious diseases ; viruses living in cells can trigger cancers / immune reactions can trigger allergies such as asthma ;	Allow: physical ill health can lead to depression / other mental illness.	2	AO1 4.2.2.7
12.2	180 ÷ 1500 × 100 ; 12 % ;		2	AO2 MS 1c
12.3	60 ÷ 2500 × 100 = 2.4 % ; 12 ÷ 2.4 ; 5 (times) ;		3	AO2 MS 1c
12.4	smoking ; lung disease / cancer / CVD ; alcohol ; liver disease / brain damage ;	Allow: carcinogens (e.g. ionising radiation) → cancer; obesity → type 2 diabetes; poor diet → CVD; any two risk factor + disease pairs.	4	AO1 4.2.2.7
12.5	benign tumours grow in one place / stay within a membrane ; malignant tumours invade neighbouring tissues ; and spread to other parts of the body in the blood (to form secondary tumours) ;		3	AO1 4.2.2.8
12.6	(a) loss of income / quality of life / cannot work ; (a) cost of treatment / need for care ; (b) cost of treating patients ; (b) fewer people working / loss of tax / more staff needed ;		4	AO3 4.2.2.7
13.1	A: (waxy) cuticle / upper epidermis ; B: palisade (mesophyll) ; C: spongy (mesophyll) ; D: lower epidermis ; E: guard cell ; F: phloem ;	One mark each. Allow xylem for F only if the student identifies the tube as xylem.	6	AO1 4.2.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
13.2	contains many chloroplasts ; near the top of the leaf so it absorbs the maximum light ; tall / thin cells packed together increase light absorption ;		3	AO2 4.2.3.1
13.3	has air spaces between the cells ; so carbon dioxide can diffuse to cells (and oxygen out) ;		2	AO2 4.2.3.1
13.4	so light can pass through to the palisade (cells) / to the chloroplasts		1	AO2 4.2.3.1
13.5	xylem ; made of dead cells / hollow tubes / strengthened with lignin ;		2	AO1 4.2.3.1
14.1	guard cells		1	AO1 4.2.3.2
14.2	guard cells take in water / become turgid (by osmosis) ; they change shape / bend, opening the stoma ;		2	AO2 4.2.3.2
14.3	no light so no photosynthesis ; so carbon dioxide is not needed / water is saved ;		2	AO2 4.2.3.2
14.4	loss of water vapour from (the surface of) the leaves ; by evaporation and diffusion through the stomata ;		2	AO1 4.2.3.2
14.5	volume = $3.14 \times 0.5^2 \times 36 = 28.3$ mm^3 ; (allow 28.26) ; $28.3 \div 6$; 4.7 mm^3 per minute ;	Allow 4.71.	4	AO2 4.2.3.2 / MS 5c
14.6	(a) stomata open wider ; more water vapour lost / faster transpiration ; (b) reduces the concentration gradient of water vapour ; so transpiration is slower ;		4	AO2 4.2.3.2
14.7	water enters root hair cells by osmosis ; moves up through the xylem ; evaporates from the mesophyll and diffuses out of the stomata ;		3	AO1 4.2.3.2
15.1	xylem: water and mineral ions ; xylem: made from dead cells / hollow ; phloem: sucrose / dissolved sugars ; phloem: living cells with pores in end walls (sieve plates) ;		4	AO1 4.2.3.3
15.2	movement of dissolved sugars / sucrose ; from the leaves to the rest of the plant (growing regions and storage organs) ;		2	AO1 4.2.3.3
15.3	sugars go up to growing shoots and down to roots / storage organs ; as the plant's needs change (sources and sinks) ;		2	AO3 4.2.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
15.4	sugars cannot move down past the ring ; so they build up above the ring ;		2	AO3 4.2.3.3
16.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • right side pumps deoxygenated blood to the lungs; left side pumps oxygenated blood around the body • blood passes through the heart twice in each circuit, increasing its pressure after the lungs • left ventricle has a thicker muscle wall than the right; it pumps blood at higher pressure to the whole body • valves between atria and ventricles and in the arteries stop backflow; blood flows in one direction • arteries have thick elastic walls to withstand pressure; veins have valves; capillaries one cell thick • link to exchange: low pressure to the lungs prevents damage to capillaries; efficient gas exchange; correct sequence of chambers		6	AO1 4.2.2.4 / 4.2.2.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
16.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • the stomach produces hydrochloric acid and protease; it churns food into a liquid • food leaving the stomach is acidic; enzymes in the small intestine need an alkaline pH • the liver produces bile which is stored in the gall bladder and released into the small intestine • bile neutralises the acid and emulsifies fat to give a larger surface area for lipase • the pancreas releases amylase, protease and lipase into the small intestine • products are absorbed into the blood: small soluble molecules (glucose, amino acids, fatty acids, glycerol) cross the villi		6	AO2 4.2.2.1–4.2.2.3
16.3	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • statins reduce cholesterol and slow the build-up of fatty material; tablets; side effects; must be taken long term • a stent keeps a coronary artery open; quick recovery; low risk but only helps one blockage; may need to be repeated • a transplant replaces a damaged heart; major surgery; donor shortage; risk of rejection; immunosuppressant drugs needed • a valve replacement or artificial heart can be used as a bridge • lifestyle change (no smoking, diet, exercise) reduces risk without surgery • balanced judgement, e.g. statins / stent for early or moderate disease, transplant only when the heart has failed		6	AO3 4.2.2.6

Where marks are lost in Topic 4.2

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Enzymes:** name the substrate and product (amylase: starch → maltose; protease: protein → amino acids; lipase: fat → fatty acids + glycerol). Say 'denatured' and 'active site changes shape' — never 'the enzyme dies'.
- ☐ **Enzyme graphs:** optimum = highest point of the curve. Use data from the graph with units in your answer.
- ☐ **Bile** is made in the liver, stored in the gall bladder, is alkaline and emulsifies fat. It is **not** an enzyme.
- ☐ **Food tests:** Benedict's → brick red (heat), iodine → blue-black, biuret → purple, ethanol → cloudy white. Always give the colour **before** and **after**.
- ☐ **Rate calculations:** rate = 1000 ÷ time. A shorter time means a faster rate. Control temperature, volumes and concentrations.
- ☐ **Heart:** learn the route: vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta. The left ventricle is thicker because it pumps to the whole body.
- ☐ **Blood vessels:** artery = thick walls + elastic fibres (high pressure); vein = valves (low pressure); capillary = one cell thick (short diffusion distance).
- ☐ **Correlation ≠ cause.** A graph showing smoking and CHD shows a link; you need a mechanism or controlled evidence to say it causes it.
- ☐ **Potometer:** volume = $\pi r^2 \times$ distance moved, then divide by time. Note the units (mm^3 per minute).
- ☐ **Transpiration:** more light, higher temperature, more wind and lower humidity all increase the rate. Use 'concentration gradient of water vapour' in answers.
- ☐ **Xylem** carries water and mineral ions up (dead cells, lignin). **Phloem** carries sucrose in both directions (living cells) — translocation.